4. Fertilization

Goals:

At the end of this lecture every student should be acquired complete knowledge about fertilization process.

Objectives:

At the end of this lecture every student should be:

- ☞ Define the new medical terms.
- ☞ Explain process of human conception (ovulation and ejaculation).

- ☞ Discuss the implantation and nourishment of the embryo before development of the placenta
- ☞ Describe normal prenatal development from conception through birth.

Outline:-

1. Introduction.
2. Objectives.
3. Definition of fertilization.
4. Definitions of new medical terms.
5. Process of fertilization.
6. Development of embryo.
7. Development of the fetus trough the nine months.

Introduction:-

A basic understanding of conception and prenatal development helps the nurse to provide care to parents during normal child bearing and better understand problems such as infertility and birth defects.

Definition of Fertilization:

Is successful of the sperm to penetrate the membrane surrounding the ovum in the ampulla (**outer third part of fallopian tube).**

Definition of new terms

1. **Gametogenesis:**

   Is the development of ova in the women **(Oogenesis**) and sperm in the man **(Spermatogenesis**), production of gametes requires a different process **than** formation of somatic cells.

2. **Oogenesis:**

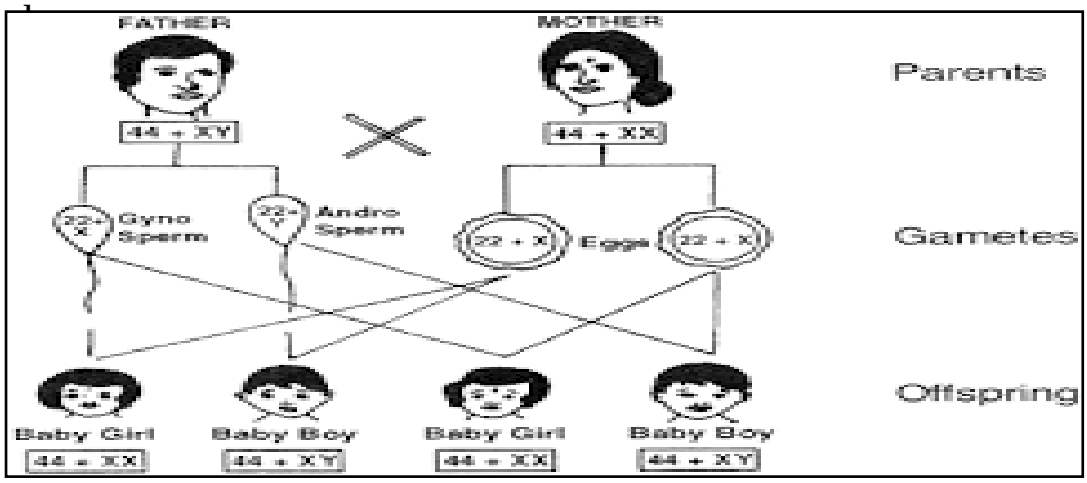
   It is the formation of **female gametes** within the ovary**.** Oogenesis begins during **prenatal life** when primitive ova multiply by mitosis, like all other cell. Each oogonium contains 46 chromosomes (22 pairs of autosomes and a pair of X chromosomes).

3. **Spermatogenesis:**

   Begins during **puberty** in male, primitive sperm cells develop during the prenatal period and begin multiplying by mitosis during puberty.

4. Sex determination

- The gamete from male determines the sex of the new baby. Each mature sperm contains 23 chromosomes (22 autosomes and one sex chromosome either X or Y).
- If an X bearing spermatozoon fertilized the ovum the baby become a girl.
- If a Y bearing spermatozoon fertilized the ovum the baby



a boy.

Fertilization Process:

This is process following ovulation:

1. The ovum which is **0.15 mm** in diameter passing into the fallopian tube and is moved along towards the uterus by the **cilia** and the **peristalsis** of the tube.
2. At in the intercourse about 60 - 150 million sperms are deposited in the posterior fornix of the vagina. Those that reach the alkaline cervical mucus propel themselves towards the fallopian tubes.
3. Some sperms are reaching to the fallopian tube where they meet the ovum in the **ampulla**. The **remainder** are **destroyed** by the acid medium of the vagina or died during the journey.
4. The mature sperm is able to release enzymes which allow them to penetrate the ovum (just only one sperm enters the ovum) and the nuclei of the two gametes (sperm and ovum) fuse to form the **zygote**.

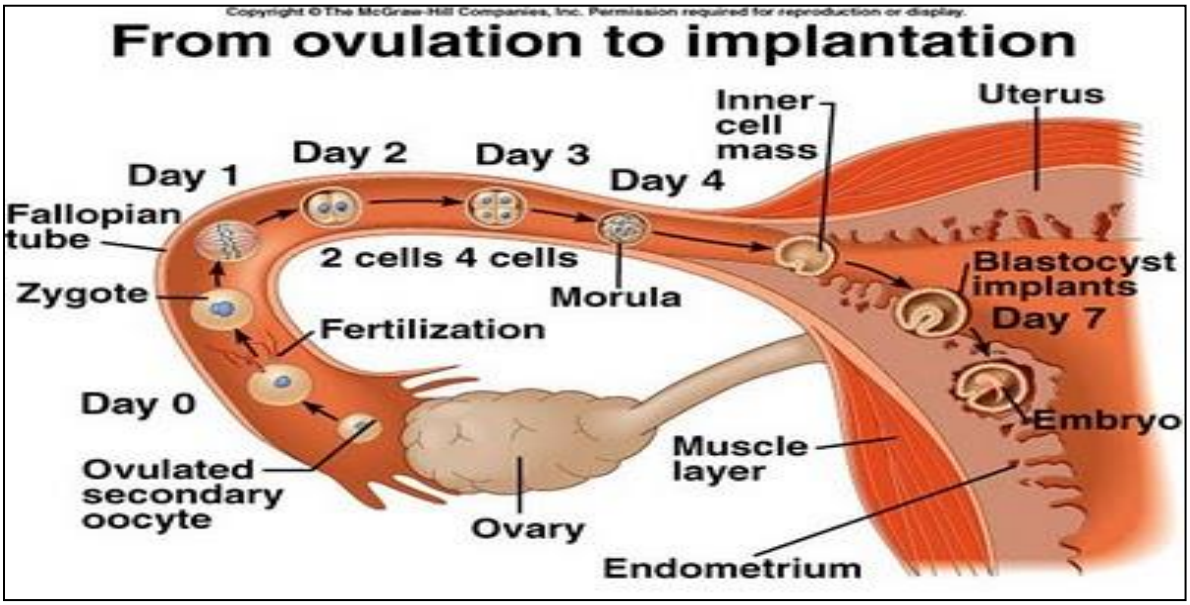
*N.B.

- The sperm reach the cervix of the uterus within 90 seconds after deposition and reach outer end of the fallopian tube in 5 minutes.
- Neither sperm nor ovum can survive for longer than 2days (for ovum) and 3 days (for sperm).

Development process of the fertilized ovum:

- When the **ovum** has been fertilized then called **zygot**, it continuo it is way through the uterine tube ***until reach to the uterus*** and that occur between **3 - 4 days** after ***fertilization***.

- During this time the divides into **2 cells, then into 4 then 8 to 16** cells then known as the **morula**. (These divisions start after 24 hours of fertilization, **and occur nearly every 12 hours**).
- After that cavity appears in the **morula** which now becomes known as the **blastocyst.**



Development of blastocyst

1. Around the outside of the blastocyst **t**here is a signale layer of cells known as the ***trophoblast*** which ***invade the uterin wall*** **and will form the placenta and chorion**
2. ile the remaining cells are clumped together at one end forming the ***inner cell mass*****.** Which ***form all tissues of the embryo***

A - The trophoblast differentiates into 2 layers.

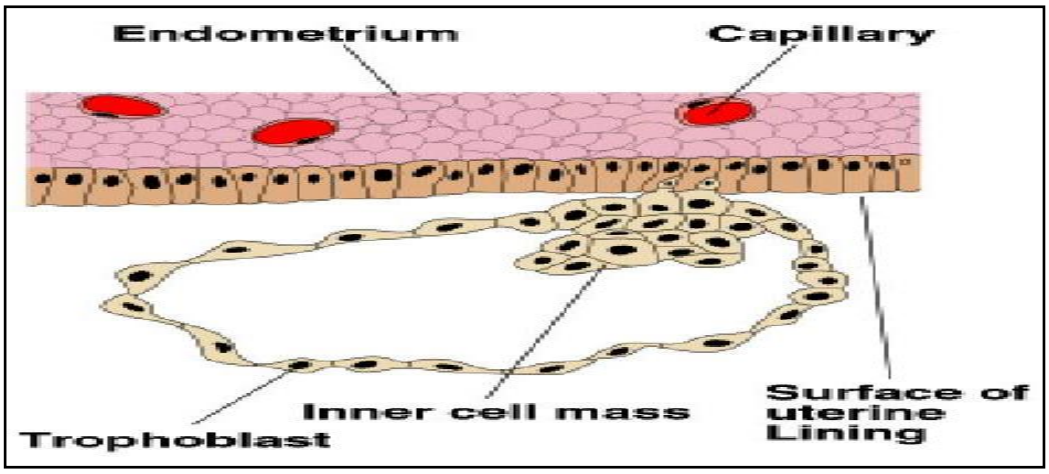
1. **The outer syncitiotrophoblast:**

It is composed of nucleated protoplasm which is capable of breaking down tissue as process of embedding it erodes the walls of maternal blood vessels of the decidua,

2. **The cytotrophoblast(Release HCG).**

- ☞ It is a well-defined single layer of cells which **produces** a hormone known as human chorionic gonadotropin hormone **(HCG).**
- ☞ This hormone is responsible for ***informing*** corpus luteum **that pregnancy has begun.**
- ☞ A third layer of **mesoderm** appears inner cytotrophoblast
- ☞ The **trophoblast** and the lining **mesoderm** together form the ***chorion*****,** and connect the inner cell mass to the chorion and will form the umbilical cord later on.
- ☞ The corpus luteum continues to produce estrogen and progesterone. Progesterone maintains the integrity of the decidua **so** that ***shedding*** does not occur.

☞ Blastocyst is completely **implanted** into decidua by the **12th – 14th** day after ovulation. The point of implantation is usually high in the uterus.



B. The inner cell mass

Will form Fetus, Amnion fluid and umbilical cord

The fetus is formed from the inner cell mass which differentiates into three layers.

- Ectoderm.
- Mesoderm.
- Endoderm.

1. **The ectoderm:**

   Which forms the skin and nervous system and the epithelium covering the outer body surface.

2. **The mesoderm:**

   Which form this forms all of the muscle tissue and the connective tissue of the body.

3. **The endoderm:**

That form mucous membranes and the glands, stomach, the colon, the liver, the pancreas, the urinary bladder, the lining of the urethra, the epithelial parts of trachea, the lungs, the pharynx, the thyroid, the parathyroid, and the intestines.

Developing stages:

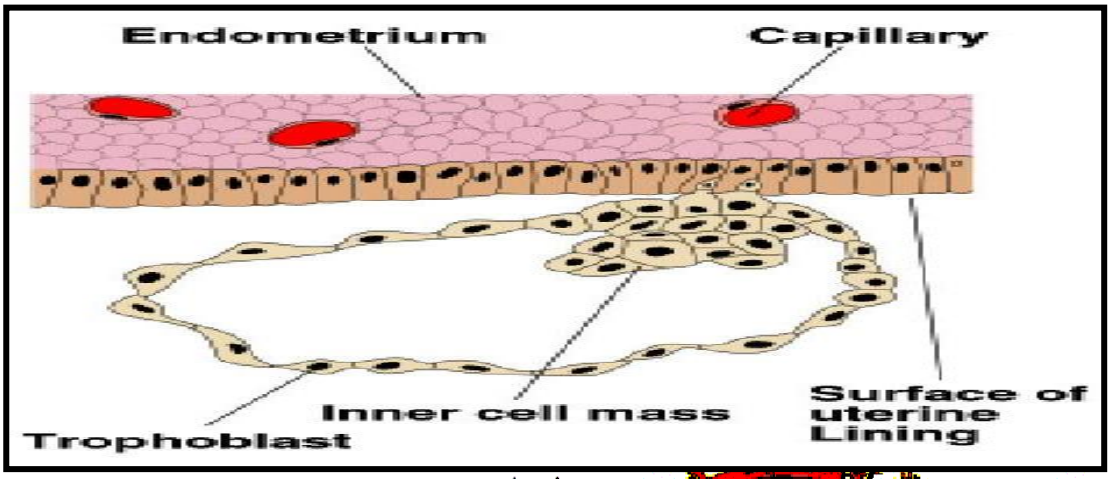
1- Ovum stage (from conception to implantation).

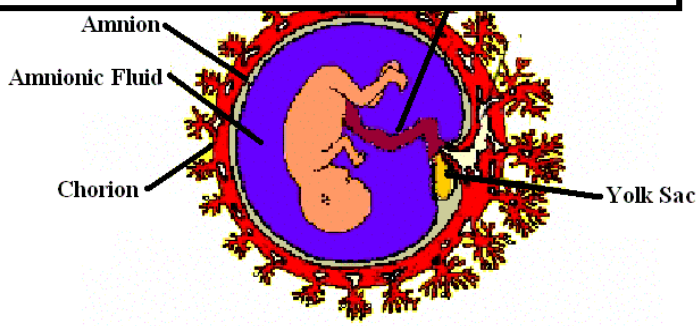
2- Embryonic stage (from 3rd week to 8th week).

3- Fetal stage (from 8th week to delivery).

Implantation (*nidation*)

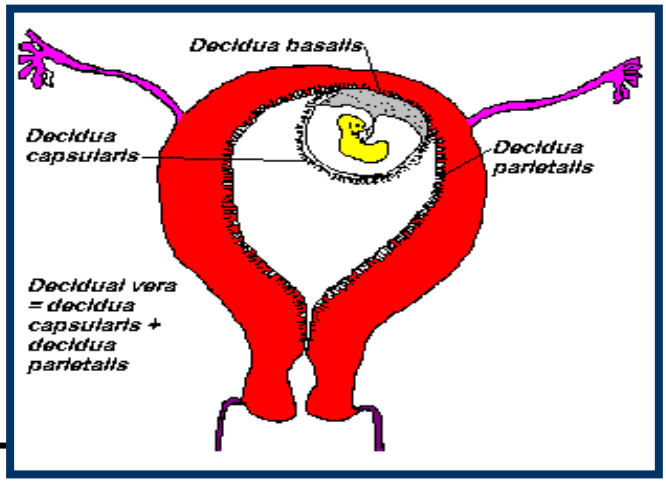
- ❖Once pregnancy occurs the endometrium of pregnant women is called **Decidua**.
- ❖From the time of conception **the increased secretion of estrogens** causes the ***endometrium to grow*** to ***four times*** it is non- pregnant thickness.
- ❖ The **corpus luteum** also **produces** large amounts of **progesterone** which **stimulate** the secretory activity of the

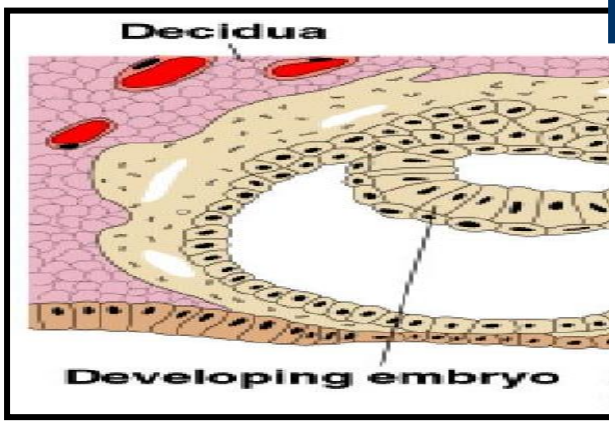




❖ **Endometrial glands and increase the size of the blood vessels.**

This accounts for the soft, vascular, spongy bed in which the fertilized ovum implants.





Development of Decidua

❖**There are three layers are found:**

1. **Decidua Basalis:** Immediately **underlying** (underneath) the blastocyst.
2. **Decidua Capsularis:** It surrounds the blastocyst as it completely embeds.
3. **Decidua Vera (Partialis):** The remaining part of the decidua which lining the rest of the uterine cavity.

The decidua has the following functions:

1. It is the site of implantation.

2. It resists more invasion of the trophoblast.

3. It nourishes the early implanted ovum by its glycogen and lipid contents.

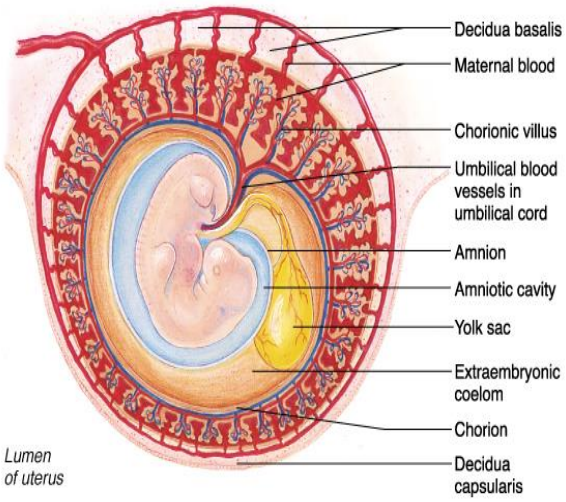
4. It shares in the formation of the placenta.

The amniotic fluid (liquor amnii):

It is a clear, pale, slightly alkaline fluid, and about one liter (from 0.5- to 1.5 liter).

1. Amniotic fluid composed of:

- Water (98-99%). - Albumen.
- Sodium chloride. - Small amounts of sugars.
- Urea. - Uric acid. - Creatinine -Ammonia.
- Enzymes and hormones e.g. estrogen.



(d) 4½-week embryo

Copyright © 2001 Benjamin Cummings, an imprint of Addison Wesley Longman, Inc.

Suspended in it:

- Lanugo hair - Vernex caseosa.
- Epithelial cells. - Sometimes meconium.

2. Origin of Amniotic Fluid :

- Transudation from fetal circulation.
- Active secretion from the amniotic epithelium.
- The fetal urine also contributes.
- Transudation from the maternal circulation.

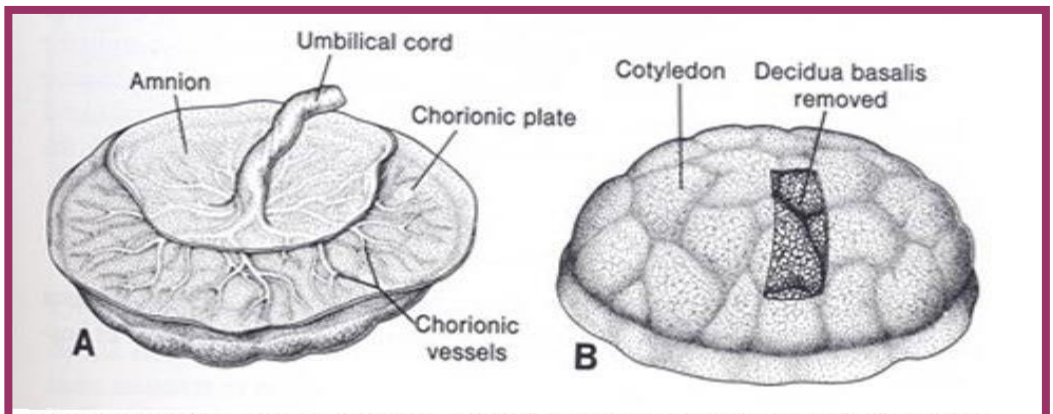
- Both the fetal and maternal origins may contribute, but probably the fetal side contributes more.

3. Function of Amniotic fluid :

- Protection of the fetus as trauma.
- Medium for free movement of the fetus.
- Keeps the fetal temperature constant.
- Nutrition as it is swallowed by the fetus.
- Medium for fetal excretion.
- The bag of membranes containing the amniotic fluid helps to dilate the cervix during labor.
- After rupture of membranes it acts as antiseptic fluid for the birth canal.

Placental development and function:-

- Placenta is a flat disk-shaped structure formed from the **chorion**, **chorionic villi** and **decidua basalis**.
- It contains 15 to 20 subdivisions called cotyledons.
- Weight of placenta is 450 : 600 gm. , measured from 10 : 25.5 cm. in diameter and is 2.5 : 3 cm. thick at full term
- The placenta has a rough texture appears red on the maternal surface and shiny and gray on the fetal surface.



A full term placenta A. Fetal side. The chorionic plate and umbilical cord are covered by amnion. B. Maternal side showing the cotyledons. In one area the decidua has been removed.

Function of the placenta

The function of placenta start from 3rd months of pregnancy until the birth of the fetus.

Pulmonary exchange of gases doesn't take place in the uterus; the fetus must obtain oxygen and excrete carbon dioxide through the placenta. The oxygen from mother's hemoglobin passes into the fetal blood by simple diffusion and similarly the fetus gives off carbon dio4ideinto the maternal blood

2. Nutrition:

the fetus needs the same nutrients as anyone (amino acids are required for body building, glucose for energy, calcium and phosphorus from bones and teeth, iron and other minerals from blood), food for the fetus derives from mother's food and has already been broken down into simple forms by the time it reaches the placental site.

3. Storage:

The placenta metabolises glucose, stores it in the form of glycogen and reconverts it to glucose as required. The placenta can also store iron and the fat soluble vitamins.

4. Excretion:

The main substances excreted from the fetus are carbon dioxide and bilirubin.

5. Protection: The placenta provides a limited barrier to infection with the exception of syphilis and the tubercle bacillus.

Foetal Membranes

Definition:

- The foetal membranes are the amnion and chorion
- The foetal membranes cover the foetus from the earliest stage until term.
- The first membrane to form is the chorion; later the amnion is formed.
- At birth they are seen on the surface of the placenta.

Amnion

- Membrane which surrounds embryo to form the amniotic cavity and produces amniotic fluid.
- Formed by the epiblast of the inner cell mass
- The amnion is a membranous sac which surrounds and protects the embryo
- Formed by the end of second week
- Contains liquor amnii
- Surrounds the foetus. Foetus floats in the liquor amnii
- Amnion enlarges and fills the whole uterus obliterating the chorionic cavity
- Amnion and chorion fuse later
- Amnion breaks and discharges the liquor at the time of delivery

Chorion

- Chorion is formed by the trophoblast
- Outermost embryonic membrane which forms the placenta & produces human chorionic gonadotropin.
- Chorion contains the chorionic cavity (extraembryonic cavity)
- The expanding amniontic cavity fills the chorionic cavity and completely obliterates it; the amniotic membrane fusing with the chorion. Amniochorionic membrane is formed
- The part of the chorion at the embryonic pole of the human embryo is called the chorion frondosum. The part away from the embryonic pole is called chorion laeve.
- The chorion at or near the placenta contains vessels of the placenta

The Umbilical Cord Formation

•The umbilical cord is formed from the mesodermal tissues, remnant of the yolk sac, amnion and chorion.

•The umbilical cord is about **2cm** in thickness and is about **50-53cm** in length at term. It contains:

- One vein which carrying oxygenated blood from the placental villi to the fetus.
- Two arteries which carrying deoxygenated blood from the fetus back to the placental villi.

- The cord is arised from the center of the fetal surface of the placenta.
- The bulk of the cord is a gelatinous mucopolysaccharide termed Wharton's Jelly which gives the cord its body and prevents pressure on the vein and arteries. The outer surface is covered with amniotic membrane.

Function of the cord:

- The cord transport oxygen and nutrients to the fetus from the placenta and to return waste products from the fetus to the placenta.

*N.B. The rate of the blood flow through the umbilical cord is rapid to ensure the continuous and rapid supply of the oxygen and nutrients.

*N.B. The umbilical cord has no nerve supply; it can cut at birth without any discomfort to infant or mother.